# Lab Exercise: DOM Manipulation with JavaScript

**Objective:** To practice DOM manipulation and events handling using JavaScript by interacting with a provided static HTML and CSS template.

### Initial Setup

Run the followingHTML code using Live Server.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Person BMI App</title>

</head>
</head>
<body>
  <div class="container">
    <h2>BMI Calculator</h2>
    <label for="name">Name:</label>
    <input type="text" id="name" placeholder="Enter your name"><br>
    <label for="yob">Year Born:</label>
    <input type="text" id="yob" placeholder="Enter year born ie 1999"><br>

    <label for="weight">Weight (kg):</label>
    <input type="number" id="weight" placeholder="Enter your weight"><br>
    <label for="height">Height (m):</label>
    <input type="number" id="height" placeholder="Enter your height"><br>
    <button id="calculateBMI">Calculate BMI</button><br>
    <p id="result"></p>
    <p id="category"></p>
  </div>
  <script>
      // write all js code here
  </script>
</body>
</html>
```

## Exercise 1: Accessing Elements via DOM

### Task 1.1: Select Elements by ID

1. Use `document.getElementById()` to select the Name input field (`<input id='name'>`) and console log it.

Expected Output: You should see the input element in the console.

Javascript code:

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### Task 1.2: Select Elements by Class

2. Select the button(`<button id='calculateBMI'>`) using `document.getElementById()` and console log it.

4. Try selecting the BMI result paragraph (`<p id='result'>`) using `document.querySelector()` and log it.

Javascript code:

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## Exercise 2: Modifying Element Content

### Task 2.1: Change Text Content

1. Use `document.getElementById()` to select the **result paragraph** and change its text content to 'BMI Calculation Result' using `textContent` or `innerText` or `innerHTML`

2. Verify the change by inspecting the webpage or checking the result in the console.

Javascript code:

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### Task 2.2: Change the Style of the Heading using DOM

3. Use JavaScript to Change the style of the <h2> header to make it bold, larger, and maroon-colored.

Javascript code:

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## Exercise 3: Handling Button Click Event

### Task 3.1: Add Event Listener

1. Use `document.querySelector()` to select the button (`<button id='calculateBMI'>`).
2. Add an event listener to the button so that when it is clicked, a message 'Button Clicked!' is logged to the console.

Javascript code:

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## Exercise 4: Getting Input Values

### Task 4.1: Get Value from Input Fields

1. After the button click, retrieve the value from the name input field using `value` and log it to the console.
2. Similarly, retrieve and log the weight and height input values.

Javascript code:

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## Exercise 5: Calculating and Displaying BMI

### Task 5.1: Perform BMI Calculation

1. Calculate the BMI using the formula:

    BMI = weight / (height * height)

2. Display the calculated BMI (i)in the console.log, and (ii) in the result paragraph.

Javascript code:

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## Exercise 6: Categorizing BMI

### Task 6.1: Categorize BMI

1. Based on the BMI value calculated in Task 5, classify/assign the user's BMI into one of the following categories:

- Underweight: BMI < 18.5
- Normal weight: 18.5 <= BMI < 24.9
- Overweight: 25 <= BMI < 29.9
- Obese: BMI >= 30

2. Display the category below the BMI value.

Javascript code:

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## Exercise 7: Adding Input Validation

### Task 7.1: Input Validation

1. Before performing the BMI calculation, check whether:
   - The name input is not empty.
   - The weight and height inputs contain valid numbers greater than zero.
2. If any input is invalid, display a warning message in the result paragraph.
3. If all inputs are valid, proceed with the BMI calculation.

Javascript code:

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## Exercise 8: Clear the Input Fields

### Task 8.1: Reset the Fields

1. Add a 'Clear' button that clears all input fields and the result when clicked.

Javascript code:

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-end of exercise-